

Ahmad Hossein Yazdani

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Research interests

I'm keen on doing research on a variety of aspects in computer systems, especially in I/O in distributed systems, cloud computing and High Performance Computing. In particular, my current research includes finding an efficient design for **Systems for ML (Especially LLMs)**, as well as employing **ML (And LLMs) models to optimize Systems**. Besides, I would like to conduct research on some hot areas like **Adapting distributed applications to an environment containing persistent memories**, **GPU scheduling of distributed applications**, in addition to **Software Hardware co-design** to optimize serverless computing environments

Education

- 2020–present **PhD, Computer Science**, *Virginia Polytechnic Institute and State University (Virginia Tech)*, Blacksburg, VA, US.
Advisor: Dr Ali Butt, **GPA:** 3.92
- May 2025 **Masters of Computer Science**, *Virginia Polytechnic Institute and State University (Virginia Tech)*, Blacksburg, VA, US.
Advisor: Dr Ali Butt, **GPA:** 3.92
- 2015–2020 : **Bachelor of Computer Software Engineering**, *University of Tehran*, Tehran, Iran.
GPA: 3.2

Conference & Workshop publications

- [ICDCN'25] **Yazdani, Ahmad Hossein**, Arnab K. Paul, Ahmad Maroof Karimi, Feiyi Wang, and Ali Butt. User-based i/o profiling for leadership scale hpc workloads. In *Proceedings of the 26th International Conference on Distributed Computing and Networking*, ICDCN '25, page 181–190, New York, NY, USA, Jan. 2025. Association for Computing Machinery. doi:10.1145/3700838.3700865.
- [**FAST'23★**] Redwan Ibne Seraj Khan, **Yazdani, Ahmad Hossein**, Yuqi Fu, Arnab K Paul, Bo Ji, Xun Jian, Yue Cheng, and Ali R Butt. Shade: Enable fundamental cacheability for distributed deep learning training. In *Proceedings of the 21th USENIX Conference on File and Storage Technologies*, page 14, Santa Clara, CA, US, Feb. 2023. USENIX Association. URL <https://www.usenix.org/conference/fast23/presentation/khan>.

★ Top-tier venue

Work Experience

August, 2020 – present **Research Assistant at Distributed System and Storage Lab, Virginia Tech.**

Advisor : Dr. Ali Butt, *Professor, Department of Computer Science, Virginia Tech*

- Contributed to Metis project ongoing which is about improving the cachability of the deep learning workloads
- Led a collaborative research with Analytics & AI Methods at Scale Group at Oak Ridge National Laboratory (ORNL) on analytically recognizing the behavior of the users and jobs submitted to HPC systems to improve the I/O efficiency of the HPC systems.

- Leading a collaborative research with Analytics & AI Methods at Scale Group at Oak Ridge National Laboratory (ORNL) and Lawrence Berkeley National Laboratory aiming to study and resolve the memory and I/O contention between the interfering training/inference jobs for large AI models (Currently LLMs) in HPC in collaboration with Jean Luca Bez, Ahmad Maroof Karimi, Arnab Kumar Paul, Suren Byna

June,2024 – ***Student Assistant at NERSC, Lawrence Berkeley National Laboratory (LBNL), internship.***
August,2024

Mentors: Stephen Simms, Lisa Gerhardt, Jean Luca Bez

- I investigated the causes of I/O hotspots in HPC applications and analyzed common performance issues. Specifically, I examined Drishti, an HPC I/O recommendation tool, and found it generates many false positive warnings. In future work, I plan to address these inaccuracies, enhance Drishti's ability to provide more reliable I/O optimization recommendations, and improve its capacity to predict job performance based on suggested configurations.

June,2023 – ***Student Assistant at Lawrence Berkeley National Laboratory (LBNL), internship.***
August,2023

Mentors: Suren Byna, Jean Luca Bez

- Continued my research on characterizing the sources of I/O performance variation in HPC, and striving to alleviate the I/O performance variability.
- Presented a poster outlining my findings on the potentials for introducing I/O interference as one cause of variability
- Continuing my efforts to mitigate I/O interference in HPC systems for LLM fine-tuning/inference.

June,2021 – ***Internship at Oak Ridge National Laboratory, Analytics & AI Methods at Scale Group.***
August,2021

Mentors: Feiyi Wang, Sarp Oral, Ahmad Maroof Karimi and Arnab Kamur Paul

- First studied the literature on I/O characterization at application level to get insights for building an application and user aware I/O scheduler
- Then collected I/O information of different users and different applications, and showed the user's behaviour affects the I/O performance quite a lot
- Then presented my work at Internship Symposium held for the interns joined the national lab in summer 2021

June,2018 – ***Summer Internship at Router lab at University of Tehran.***
August,2018

- Improved the lab's website in terms of responsiveness and SEO, helped me improve my web application front-end design skills
- Read CISCO's documentations to make the lab router's CLI identical to CISCO. Also, I was invited to participate in a project for adding QoS to data plane of the lab's routers.

Notable projects

Fall 2023 - ***LLM tensor storage offloading system.***

- Leading the project which extends SHADE for LLMs. LLMs require a huge amount of memory across tiers like File system (Lustre in this case) and RAM to store the activations generated during the encode and decode phases along with my collaborators from ORNL and LBNL.

Fall 2020 - ***SHADE: Enable Fundamental Cacheability for Distributed Deep Learning Training.***
Fall 2022

- Contributed to SHADE, a new DLT-aware caching system that detects fine-grained importance variations at per-sample level and leverages the variance to make informed caching decisions for a distributed DLT job. Mainly used Chameleon Cloud for the experiments.

- Fall 2020 ***Container layer caching evaluation for Docker.***
- Implemented and evaluated various eviction policies like LRU and LFU for Docker container layer caching as a course project for CS5204
- Spring 2020 ***Stock market price prediction using RNNs.***
- Used Recurrent Neural Networks (RNNs) to predict Iran's stock market. Specifically, to predict the opening price of the stocks for the next couple of days given a history of the open, high, low and close prices. Used Google Colab for training
- Spring 2019 ***Toorla Compiler.***
- Developed Toorla compiler in Java, a java-like compiler to ByteCode featuring Object Oriented programming concepts like class inheritance, method overriding and overloading, etc.
- Fall 2018 ***Polling website development using Django.***
- Developed a website with simple UI backed with Django featuring poll creation, closure, etc. Used ReactJS for front-end development
- Spring 2017 ***Housing website development using NodeJS/Java EE.***
- Developed a website with simple UI backed with nodeJS where the information associated with available houses are displayed. Backed with both NodeJS and Java EE. Use SQLite as the database.
- Spring 2017 ***Random boosting-based semaphore in Linux Kernel.***
- Developed an extended semaphore deep inside the Linux kernel to boost the priority of processes randomly if a process with higher priority is stuck waiting for a resource not available readily

Fellowships & Awards

- 2024 ***TCPP travel grant recipient for IPDPS24***, San Francisco, CA
- 2024 ***USENIX travel grant recipient for FAST24***, Santa Clara, CA
- 2022 ***Student Volunteer at SC22***, Dallas, TX
- 2023 ***Student Volunteer at SC23***, Denver, CO

Presentations

- 2024 ***IPDPS24***, A conference present their latest research findings in all aspects of parallel computation and distributed processing. In addition to technical sessions of submitted paper presentations. I presented a poster at the IPDPS PhD forum our recent findings on the I/O interference project accomplished in collaboration with Lawrence Berkeley National Laboratory (LBNL).
- 2022 ***MUG22***, A meeting sharing the recent advancements on MVAPICH (A library overlaying MPI), and how these improvements impact the applications. I presented a poster on my research on identification of the role of users in affecting the I/O performance of the HPC applications in collaboration with Oak Ridge National Laboratory (ORNL)

Computer skills

- Programming Languages & frameworks Python, R, C, C++, Advanced JAVA EE, JAVA SE, Tensorflow, Go, Rust, MPI, pthread, OpenMP, CUDA, Java bytecode, X86 assembly
- Machine Learning & GPU programming PyTorch, keras, scikit-learn, DeepSpeed, LLM, Computer Vision
- Systems Linux kernel programming, distributed systems, File Systems, Lustre, GPFS, Slurm, MapReduce, Redis, Ceph, NVMe, Darshan, Docker, Kubernetes

Web Technologies & Enterprise application development HTML5, PHP, JSP, Javascript, Django, nodeJS, WordPress, Apache tomcat, Maven, Junit, software testing, REST

Database SQL, MySQL, MSSQL

Notable teaching experience

Virginia Tech

Fall 2023, 2024 **CS3214: Computer Systems, head TA.**

- Served as the head TA; creating the rubrics for the assignments and coordinating the logistics.

Spring 2023 : **CS3214: Computer Systems, instructor.**

- Giving presentations to one section (75 students) in parallel with two other sections taught by Godmar Back and Dan Williams.

Fall 2022 : **CS3214: Computer Systems, instructor.**

- Giving presentations to one section (75 students) in parallel with two other sections taught by Godmar Back and Huaicheng Li.

University of Tehran

Spring 2020 : **Artificial intelligence, teaching assistant.**

- hosted project help session, created a project assignment and homework assignment, grading

Spring 2019 : **Programming Languages and Compilers, teaching assistant.**

- created and led the project course, hosted a help session for each phase of the project, grading

Referees

Dr. Ali Butt

*Professor, Department of
Computer Science
Virginia Tech*

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Dr. Jean Luca Bez

*Data Management Research Scientist
Scientific Data Division
Berkeley Lab, US*

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Dr. Arnab Kumar Paul

*Assistant Professor, Department of
Computer Science and Information Systems
BITS Pilani, K K Birla Goa Campus, India*

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Dr. Suren Byna

*Professor, Department of
Computer Science and Engineering (CSE)
The Ohio State University (OSU), US*

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Dr. Ahmad Maroof Karimi

*HPC Operational Data Scientist in
Analytics and AI Methods at Scale Group
Oak Ridge National Laboratory*

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